

Sports Club Membership & Tournament Management System

PHASE 1: CONCEPTUAL MODELING, ENTITY-RELATIONSHIP DESIGN & RELATIONAL SCHEMA (3NF)

12 Normalized Relations

Entity-Relationship Architecture

Third Normal Form (3NF)

ACID Compliance

SQL Triggers & Views

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1.0 PROJECT OVERVIEW • INTRODUCTION

Centralizing athletic operations, memberships, facilities, and competitive tournaments

📌 Background & Motivation

Modern athletic facilities and sports clubs operate across diverse disciplines (Cricket, Football, Tennis, Swimming, Badminton, Basketball). Managing these operations requires coordinating hundreds of athlete memberships, certified coaches, training facility slots, subscription fee collections, and tournament schedules.

- ▶ **Multi-Disciplinary Scope:** Unified platform managing athletes across 6+ sports simultaneously.
- ▶ **Relational Paradigm:** Eliminates legacy spreadsheet islands and paper logs with ACID-compliant storage.
- ▶ **Role-Based Access:** Dedicated privileges for System Administrators, Coaches, and Athlete Members.
- ▶ **Data Integrity & Automation:** Guaranteed atomicity in financial transactions and automated schedule overlap prevention.

⚡ Core Operational Domains

- ▶ **Athlete Profiles & Authentication:** Demographic data, SHA-256 password hashing, and membership lifecycle states (Active, Expired, Suspended).
- ▶ **Club Franchises & Tactical Rosters:** Many-to-Many team enrollments with specific positional roles (Striker, Wicketkeeper, Point Guard).
- ▶ **Multi-Tier Subscriptions & Billing:** Flexible plan tiers (Monthly, Quarterly, Gold Pass) with multi-mode payment auditing.
- ▶ **Training & Facility Scheduling:** Ground/court reservations with capacity tracking and conflict avoidance.
- ▶ **Tournament Brackets & Match Fixtures:** League/knockout match organization with automated standings computation.

2.0 PROBLEM STATEMENT • OPERATIONAL CHALLENGES

CHALLENGES

Overcoming critical anomalies and bottlenecks in traditional club administration

✖ Data Redundancy & Anomalies

Insertion & Deletion Anomalies:
In un-normalized schemas, coach contact details or sport definitions cannot be added without enrolling a member, leading to NULL values and inconsistent records.

Update Inconsistencies:
Modifying a coach's phone number across multiple spreadsheet sheets often leaves orphaned or contradictory records.

⚠ Scheduling & Venue Overlaps

Double-Booking Facilities:
Simultaneous training sessions and tournament matches scheduled at the same stadium turf or indoor arena create major operational collisions.

Sport Incompatibility:
Without database validation triggers, tournament fixtures can mistakenly pair teams registered under different sports disciplines.

≡ Financial & Roster Blindspots

Uncollected Subscription Fees:
Lack of atomic transaction linking allows athletes with expired dues to continue booking premium training sessions.

Untracked Player Roster Roles:
No historical tracking of tactical positions or membership duration across multiple clubs.

3.0 PROJECT OBJECTIVES • SYSTEM SCOPE

Architectural goals and engineering scope for the relational database system

Primary Project Objectives

- ▶ **3NF Normalization:** Decompose data into 12 atomic relations with zero transitive dependencies to guarantee mathematical data integrity.
- ▶ **Enforce Referential & Domain Constraints:** Utilize Foreign Keys, composite primary keys, and CHECK constraints for all vital fields.
- ▶ **Automated Business Integrity Triggers:** Implement procedural triggers preventing venue booking collisions and tournament team sport mismatches.
- ▶ **Sub-Millisecond Query Performance:** Pre-compile analytical views ('v_club_rosters', 'v_tournament_standings', 'v_facility_utilization') for high throughput.
- ▶ **Full-Stack Interoperability:** Seamless interface with Python HTTP backend and interactive Single-Page Application (SPA).

Functional Scope Breakdown

Subsystem	Core Responsibilities
Membership Module	Athlete registrations, demographics, login authentication, status monitoring.
Finance & Ledger	Plan subscriptions, multi-mode payment auditing, GST invoices, renewal alerts.
Club & Roster Module	Franchise management, head coach assignments, tactical role enrollment.
Facility & Training	Court/pitch reservations, practice slot capacity control, coach supervision.
Tournament & Match	Knockout & league fixture generation, scoring results, points standings.

4.0 ENTITY IDENTIFICATION • ATTRIBUTE CATALOG

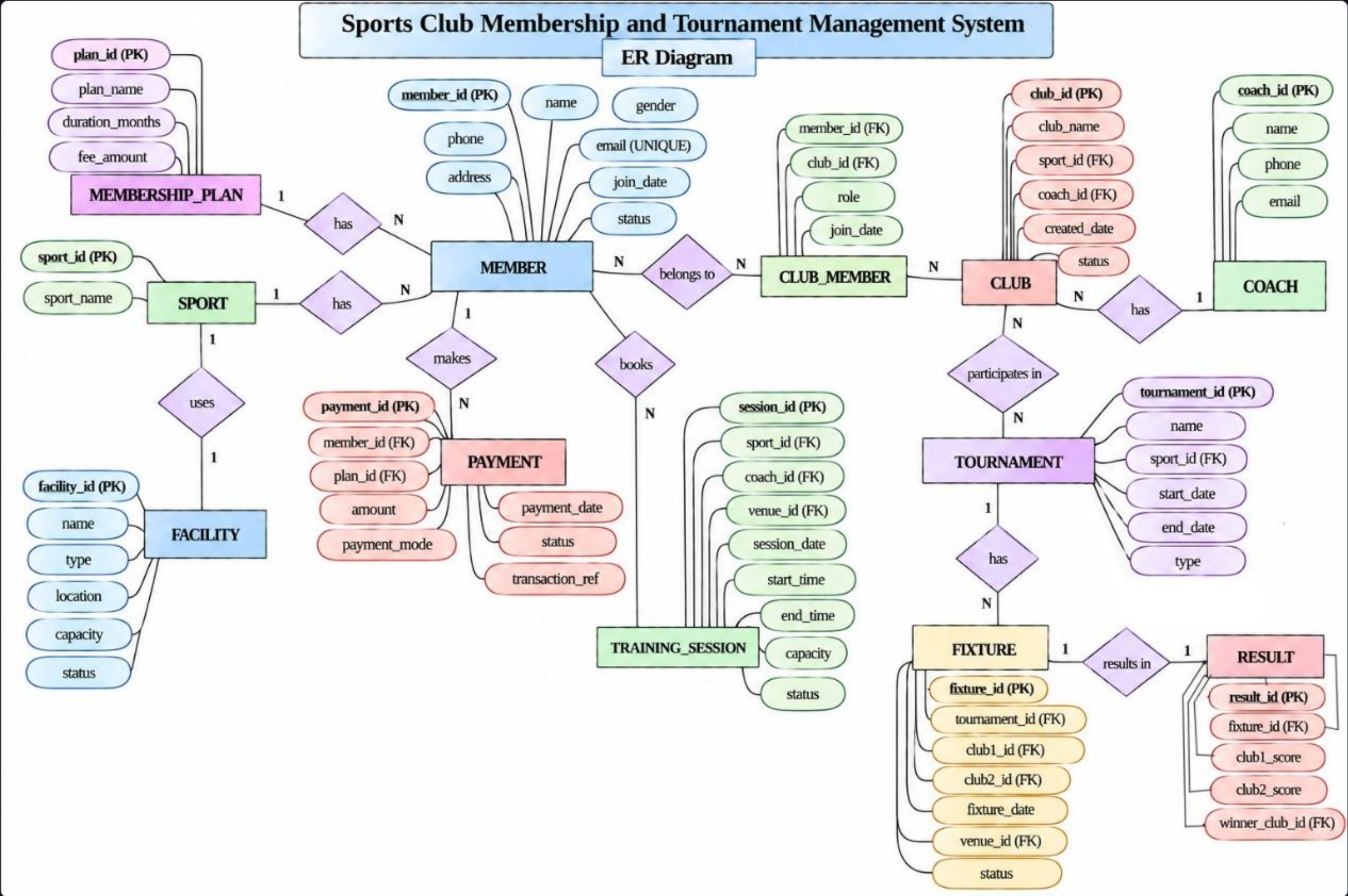
DATA MODEL

Classification of strong, weak, and associative entities with key attributes

MEMBER MEMBER_ID (PK) name gender email password phone / address join_date / status VARCHAR(100) NOT NULL VARCHAR(10) CHECK VARCHAR UNIQUE SHA-256 HASH VARCHAR / TEXT DATE / VARCHAR	CLUB & ROSTER CLUB_ID (PK) club_name sport_id coach_id status CLUB_MEMBER join_date VARCHAR(100) NOT NULL FK → SPORT FK → COACH VARCHAR CHECK (M_ID, C_ID, ROLE) DATE DEFAULT	PAYMENT & PLAN PAYMENT_ID (PK) member_id / plan_id amount payment_mode transaction_ref payment_date / status duration_months FKS → MEMBER, PLAN DECIMAL(10,2) > 0 VARCHAR CHECK VARCHAR UNIQUE DATE / VARCHAR INT IN PLAN
SPORT & COACH SPORT_ID / COACH_ID sport.sport_name coach.name coach.email coach.phone discipline_code VARCHAR UNIQUE VARCHAR(100) NOT NULL VARCHAR UNIQUE VARCHAR(20) INDEXED	FACILITY & SESSION FACILITY_ID / SESSION_ID facility.name / type facility.capacity session.venue_id coach_id / sport_id start_time / end_time VARCHAR NOT NULL INT CHECK > 0 FK → FACILITY FKS → COACH, SPORT TIME (end > start)	TOURNAMENT & FIXTURE TOURNAMENT_ID / FIXTURE_ID tournament.name / type start_date / end_date team1_id / team2_id fixture_date / venue_id match_status VARCHAR NOT NULL DATE (end ≥ start) FKS → TEAM (T1 ≠ T2) DATETIME / FK VARCHAR CHECK

5.0 ENTITY-RELATIONSHIP DIAGRAM • CONCEPTUAL MODEL

Visualizing entities, attributes, primary/foreign keys, and cardinalities



6.0 RELATIONAL SCHEMA • KEY MAPPINGS

Formal schema representation mapping primary keys, candidate keys, and foreign relationships

Formal Relational Notation (3NF)

```
MEMBER (member_id, name, gender, phone, email, password, address, join_date, status)
SPORT (sport_id, sport_name)
MEMBERSHIP_PLAN (plan_id, plan_name, duration_months, fee_amount)
FACILITY (facility_id, name, type, location, capacity, status)
COACH (coach_id, name, phone, email)
CLUB (club_id, club_name, created_date, status, sport_id, coach_id)
CLUB_MEMBER (member_id, club_id, role, join_date)
PAYMENT (payment_id, member_id, plan_id, amount, payment_mode, payment_date, status, transaction_ref)
TRAINING_SESSION (session_id, session_date, start_time, end_time, capacity, status, sport_id, coach_id, venue_id)
TEAM (team_id, team_name, created_date, status, club_id, sport_id, coach_id)
TOURNAMENT (tournament_id, name, start_date, end_date, type, sport_id)
FIXTURE (fixture_id, fixture_date, status, tournament_id, team1_id, team2_id, venue_id)
```

Cardinality & Relationship Rules

- MEMBER ↔ CLUB (N:M): An athlete can enroll in multiple clubs with distinct tactical roles; a club holds multiple athletes. Resolved by CLUB_MEMBER .
- MEMBER ↔ PAYMENT (1:N): An athlete performs multiple subscription transactions over time; each payment record belongs to exactly one athlete.
- SPORT ↔ CLUB (1:N): Each franchise is dedicated to one sport discipline; a sport hosts multiple competing clubs.
- FACILITY ↔ TRAINING_SESSION (1:N): A venue hosts multiple time slots sequentially; each session occupies one venue.
- TOURNAMENT ↔ FIXTURE (1:N): A tournament organizes a bracket of matches between competing squads (team1_id ≠ team2_id) .

7.0 CORE TABLE DEFINITIONS • DATA TYPES & RULES

Physical schema attributes, domain types, constraints, and business rules across all relations

Table Name	Primary Key	Foreign Keys	Key Attributes & Types	Integrity & Business Rules
member	MEMBER_ID	—	name (VARCHAR), gender, phone, email (VARCHAR UNIQUE), password, status	CHECK (gender IN ('Male','Female')), CHECK (status IN ('Active','Expired','Pending','Suspended','Inactive'))
membership_plan	PLAN_ID	—	plan_name (VARCHAR), duration_months (INT), fee_amount (DECIMAL)	CHECK (duration_months > 0), CHECK (fee_amount >= 0)
payment	PAYMENT_ID	MEMBER_ID, PLAN_ID	amount (DECIMAL), payment_mode, payment_date, status, transaction_ref (UNIQUE)	CHECK (amount > 0), CHECK (payment_mode IN ('UPI','Credit Card','Debit Card','Net Banking','Cash'))
sport	SPORT_ID	—	sport_name (VARCHAR UNIQUE)	Parent catalog for clubs, training sessions, and tournament categories
coach	COACH_ID	—	name (VARCHAR), phone, email (VARCHAR UNIQUE)	Certified coach profile referenced by clubs, squads, and training sessions
club	CLUB_ID	SPORT_ID, COACH_ID	club_name (VARCHAR), created_date, status	Foreign key cascade protection; coach assignment validity
club_member	(MEMBER_ID, CLUB_ID, ROLE)	MEMBER_ID, CLUB_ID	role (VARCHAR), join_date (DATE)	Composite primary key prevents duplicate role registration within same franchise
facility	FACILITY_ID	—	name, type, location, capacity (INT), status	CHECK (capacity > 0), CHECK (status IN ('Available','Booked','Under Maintenance'))
training_session	SESSION_ID	SPORT_ID, COACH_ID, VENUE_ID	session_date, start_time, end_time, capacity, status	CHECK (end_time > start_time), Automated venue overlap trigger enforcement
team	TEAM_ID	CLUB_ID, SPORT_ID, COACH_ID	team_name (VARCHAR), created_date, status	Competitive tournament squad formed under sports clubs
tournament	TOURNAMENT_ID	SPORT_ID	name (VARCHAR), start_date, end_date, type	CHECK (end_date >= start_date), CHECK (type IN ('Knockout','Round Robin','League'))
fixture	FIXTURE_ID	TOURNAMENT_ID, TEAM1_ID, TEAM2_ID, VENUE_ID	fixture_date, status	CHECK (team1_id <> team2_id), Trigger validating teams match tournament sport

Declarative Integrity Rules

- ▶ **Entity Integrity:** Primary Keys enforce uniqueness on all entities ('member_id', 'club_id', 'payment_id', 'session_id').
- ▶ **Referential Integrity:** Foreign Keys guarantee orphan-free child records with 'ON DELETE RESTRICT / CASCADE'.
- ▶ **Domain Constraints:** Check conditions restrict values (e.g., 'amount > 0', 'end_time > start_time', valid payment modes).
- ▶ **Unique Constraints:** Prevents duplicate registration of emails ('member.email', 'coach.email') and transaction reference numbers.

Procedural Trigger: Sport Validation

```
-- Ensures competing squads participate in tournament sport
CREATE TRIGGER trg_check_fixture_team_sport
BEFORE INSERT ON fixture
FOR EACH ROW
BEGIN
    SELECT CASE
        WHEN (SELECT sport_id FROM team WHERE team_id = NEW.team1_id) <>
             (SELECT sport_id FROM tournament WHERE tournament_id = NEW.tournament_id)
        THEN RAISE(ABORT, 'Error: Team 1 sport mismatch!')
        WHEN (SELECT sport_id FROM team WHERE team_id = NEW.team2_id) <>
             (SELECT sport_id FROM tournament WHERE tournament_id = NEW.tournament_id)
        THEN RAISE(ABORT, 'Error: Team 2 sport mismatch!')
    END;
END;
```

9.0 CONCLUSION • PHASE 1 SUMMARY

Mathematical validation, normalization benefits, and upcoming full-stack milestones

SUMMARY

✔ Phase 1 Accomplishments

- ▶ **Complete Conceptual Design:** Comprehensive Entity-Relationship modeling covering 12 core athletic domains.
- ▶ **Strict 3NF Normalization:** Mathematical elimination of update, insertion, and deletion anomalies.
- ▶ **Robust Constraint Architecture:** Entity, Referential, and Trigger-based integrity guards implemented.
- ▶ **Scalable Schema:** Extensible structure accommodating multi-sport complexes, tournaments, and payment simulation.

🚀 Roadmap & Next Phase Milestones

- ▶ **Python REST API Backend:** High-performance HTTP server handling authentication and query execution.
- ▶ **Dual-Role Web Portals:** Responsive Single-Page Applications for System Administrators and Athletes.
- ▶ **Smart SQL Studio:** Schema-aware Natural Language-to-SQL compiler with instant query profiling.
- ▶ **Interactive Club Management:** Real-time club enrollment, role switching, and payment processing.